

Mini Project

on

**“Real-Time Location Monitoring and Geofence Alert
System for Alzheimer’s”**

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

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under the guidance of

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**Department of Electronics and Communication Engineering
BVRIT HYDERABAD College of Engineering for Women**

(UGC Autonomous Institution | Approved by AICTE | Affiliated to JNTUH)
(NAAC Accredited – A Grade | NBA Accredited B. Tech. (EEE, ECE and CSE))

Bachupally, Hyderabad – 500090

2025-26

DECLARATION

We hereby declare that the work described in this report, entitled **“RealTime Location Monitoring and Geofence Alert System for Alzheimer’s”** which is being submitted by us in partial fulfilment for the award of the degree of **Bachelor of Technology** in the Department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women (Autonomous)**, affiliated to **Jawaharlal Nehru Technological University Hyderabad**, Kukatpally, Hyderabad – 500090 is the result of original work carried out by us under the guidance of **Dr. Nagesh Deevi Associate Professor & HOD**.

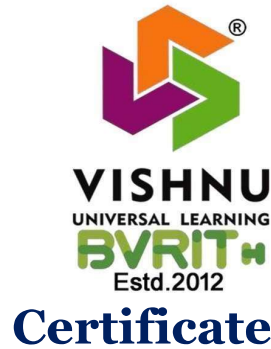
This work has not been submitted for any Degree/Diploma of this or any other institute/university to the best of our knowledge and belief.

Place: Hyderabad

Date:

Names and signatures of the students

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This is to certify that the major/mini project report, entitled “**Real-Time Location Monitoring and Geofence Alert System for Alzheimer’s**” is a record of bonafide work carried out by **Y. Jyothi – 24WH5A0404, K. Lakshmi Sri – 23WH1A0446, K. Akshaya – 23WH1A0449, Mamatha Kumari – 23WH1A0460** in partial fulfilment for the award of the degree of **Bachelor of Technology** in the department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women (Autonomous)**, affiliated to **Jawaharlal Nehru Technological University Hyderabad**, Kukatpally, Hyderabad – 500090.

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ABSTRACT

This project presents a real-time monitoring system designed for Alzheimer's patients to ensure their safety and prevent wandering incidents. The system uses an ESP32 microcontroller integrated with a Neo-6M GPS module to continuously track the patient's location. The collected location data is transmitted through Wi-Fi to a cloud platform, where it can be accessed through a web-based application by caregivers. A geofencing technique is implemented to define a safe zone around a fixed location such as the patient's home. When the patient moves beyond this predefined boundary, the system automatically generates alerts to notify caregivers. The alerts include notifications on the web interface and message-based alerts for immediate attention. The system provides real-time location tracking, low-cost implementation, and ease of use, making it suitable for continuous patient monitoring. It enhances patient safety by enabling quick response in case of emergencies.

CONTENTS

S.No	Topic	Page No.
1	Introduction	1
1.1	Introduction to Alzheimer's Monitoring System	1
1.2	Need for Real-Time Tracking	1
2	Literature Survey	2
3	System Overview	2-3
3.1	Block Diagram Description	2
3.2	Working Principle	3
4	Hardware Description	3-4
4.1	ESP32 Microcontroller	3
4.2	Neo-6M GPS Module	4
4.3	Power Supply	4
5	Software Description	4
5.1	Arduino IDE	4
5.2	Web Application	4
6	Methodology	5
6.1	Data Acquisition using GPS	5
6.2	Data Transmission using Wi-Fi	5
6.3	Geofencing Algorithm	5
7	Implementation	5-6
8	Results and Discussion	6-7
9	Advantages and Limitations	7
10	Applications	7
11	Conclusion and Future Scope	8
12	References	8-9

LIST OF FIGURES AND TABLES

S.No	Topic	Page No.
1.	Fig.1 Block Diagram	2
2.	Fig.2 ESP32 Microcontroller	3
3.	Fig.3 Neo-6M GPS Module	4
4.	Fig.4 Circuit Diagram	5
5.	Fig.5 Web Dashboard	6
6.	Fig.6 Alert	6
7.	Fig.7 SMS	6
8.	Fig.8 Serial Monitor	6

1. INTRODUCTION

Alzheimer's disease is a progressive neurological disorder that affects memory, thinking ability, and behaviour. Patients suffering from Alzheimer's often experience confusion, memory loss, and disorientation, which may cause them to wander away from safe locations. This can lead to serious safety risks and makes continuous monitoring essential.

To address this problem, a real-time monitoring system is required that can track the patient's location and alert caregivers when the patient moves beyond a safe area. This project proposes a GPS-based tracking system using an ESP32 microcontroller and a Neo-6M GPS module.

The system continuously collects the patient's location data and transmits it to a cloud platform using Wi-Fi. Caregivers can monitor the patient's position through a web application and receive alerts if the patient crosses a predefined boundary. This solution is cost-effective, reliable, and easy to use.

1.1 Introduction to Alzheimer's Monitoring System

An Alzheimer's monitoring system is designed to ensure the safety of patients by tracking their real-time location. It uses GPS technology to determine the patient's position and communication technologies such as WiFi to send this data to a remote server.

The system enables caregivers to monitor patients remotely and respond quickly in case of emergencies. It reduces the risk of patients getting lost and provides peace of mind to family members.

1.2 Need for Real-Time Tracking

Real-time tracking is essential for Alzheimer's patients due to their tendency to forget directions and wander away.

Key needs include:

- Continuous monitoring of patient location.
- Immediate detection of unsafe movement.
- Quick alert generation for caregivers.
- Improved safety and emergency response.

2. LITERATURE SURVEY

Various systems have been developed for tracking patients using GPS and communication technologies. Earlier systems mainly used GSM modules to send location information through SMS.

Recent developments in the Internet of Things (IoT) have enabled real-time monitoring using cloud platforms and web applications. These systems provide better accuracy and faster updates compared to traditional methods.

However, many existing systems are either expensive or complex to implement. The proposed system focuses on a simple and low-cost solution using ESP32 and GPS technology, making it suitable for practical use.

3. SYSTEM OVERVIEW

The system consists of hardware and software components working together to track the patient's location.

Main components include:

- GPS Module for location tracking.
- ESP32 microcontroller for data processing.
- Wi-Fi for data transmission.
- Cloud platform for data storage.
- Web application for monitoring and alerts

3.1 Block Diagram Description

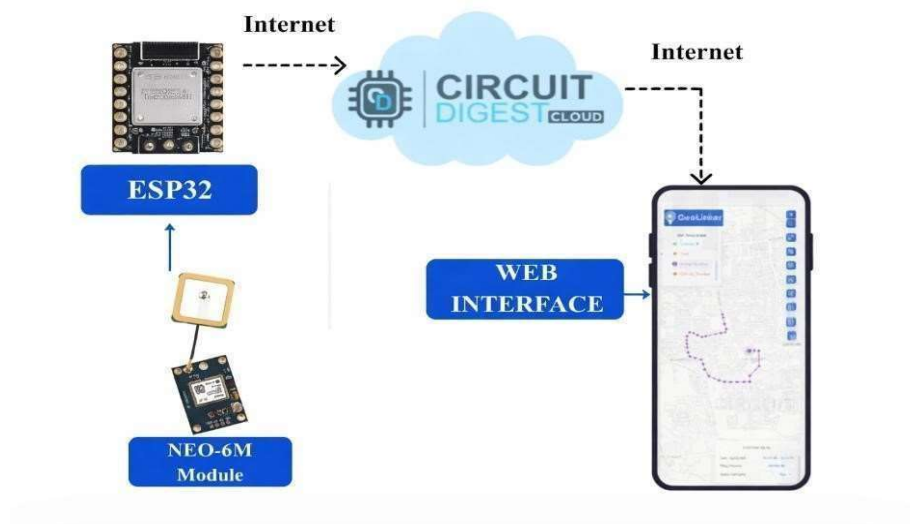


Fig.1

3.2 Working Principle

The Alzheimer's patient monitoring system operates by continuously tracking the patient's geographical location using a GPS module and transmitting this information to a cloud platform through the ESP32 microcontroller.

The Neo-6M GPS module receives signals from satellites and calculates the current position in terms of latitude and longitude. This data is sent to the ESP32 through serial communication. The ESP32 processes this data and connects to a Wi-Fi network to upload the location coordinates to the cloud server.

The system continuously calculates the distance between the patient's current location and a predefined home location. A geofencing technique is used to define a safe boundary around this location.

If the patient moves outside the safe zone, the system detects the condition and triggers alerts such as SMS notifications and cloud updates. If the patient returns within the safe zone, the system resets the alert and continues monitoring.

4. HARDWARE DESCRIPTION

4.1 ESP32 Microcontroller

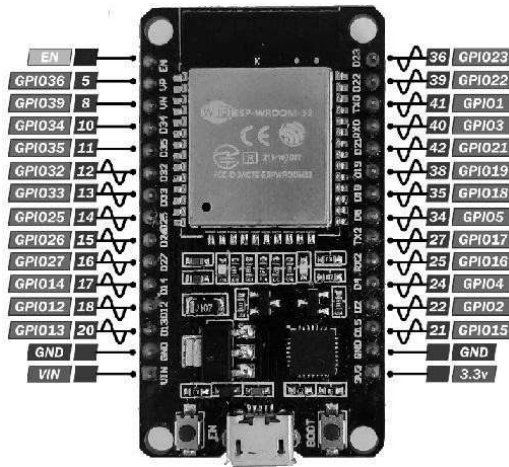


Fig.2

ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth capabilities. It is used to process GPS data and transmit it to the cloud platform. It is cost-effective and suitable for IoT applications.

4.2 Neo-6M GPS Module

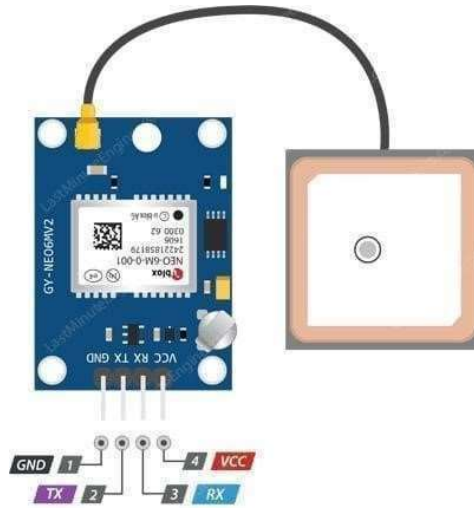


Fig.3

4.3 Power Supply

The system requires a stable power supply to operate properly. ESP32 works at 3.3V, and proper voltage regulation ensures reliable performance.

5. SOFTWARE DESCRIPTION

5.1 Arduino IDE

Arduino IDE is used to write, compile, and upload code to the ESP32. It provides an easy-to-use environment for programming.

5.2 Cloud Platform

The cloud platform is used to store GPS data sent from the ESP32. It allows real-time access to location data.

5.3 Web Application

The web application displays the patient's location and alerts caregivers when necessary. It provides a user-friendly interface.

6. METHODOLOGY

6.1 Data Acquisition using GPS

The GPS module continuously collects the patient's location data and sends it to the ESP32.

6.2 Data Transmission using Wi-Fi

The ESP32 connects to Wi-Fi and sends location data to the cloud platform.

6.3 Geofencing Algorithm

Geofencing defines a virtual boundary around a safe location.

Steps involved:

- Define safe zone using coordinates and radius.
- Get current location from GPS.
- Calculate distance using Haversine formula.
- Compare with radius.
- Generate alert if boundary is crossed.
- Reset when patient returns.

7. IMPLEMENTATION

The system is implemented using both hardware and software components.

Hardware Implementation:

GPS TX → ESP32 RX (GPIO 16) GPS

RX → ESP32 TX (GPIO 17)

VCC → 3.3V

GND → GND

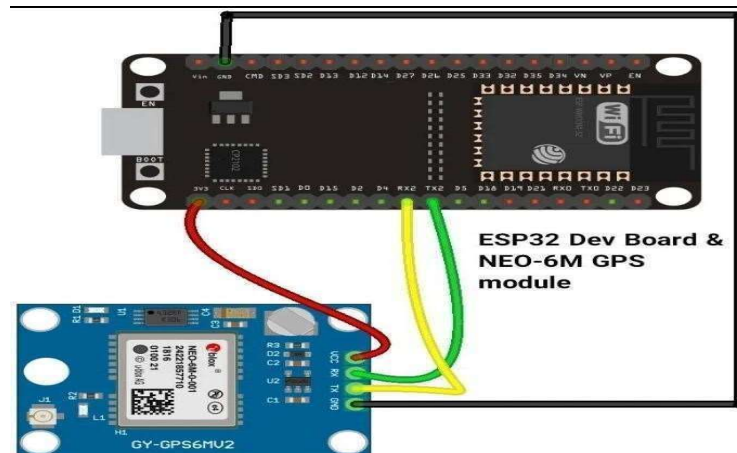


Fig.4

Software Implementation:

- Code written in Arduino IDE.
- Uses TinyGPSPlus library.
- Sends data to cloud.
- Checks geofence condition.
- Sends SMS alert.

8. RESULTS AND DISCUSSION

Results:

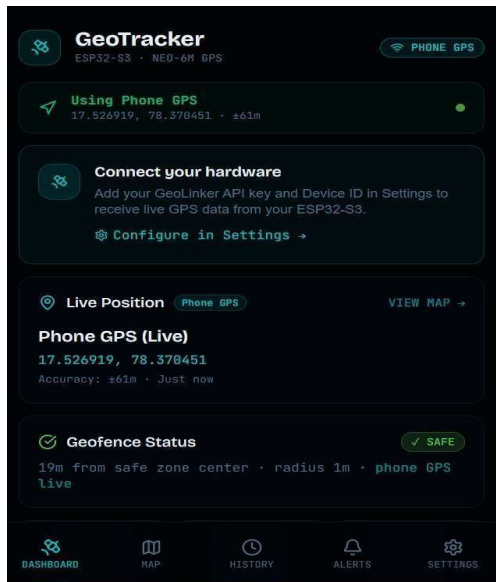


Fig.5

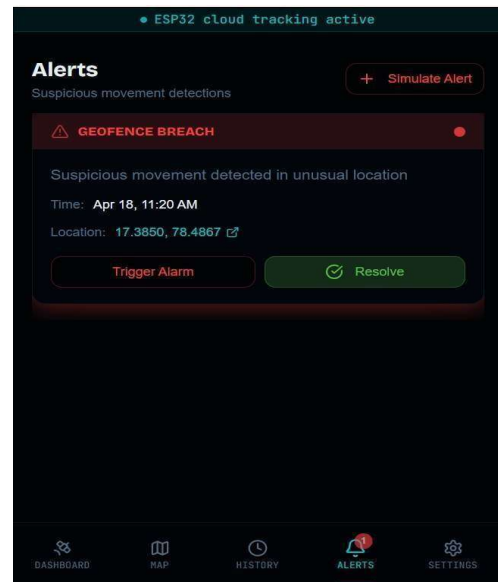


Fig.6

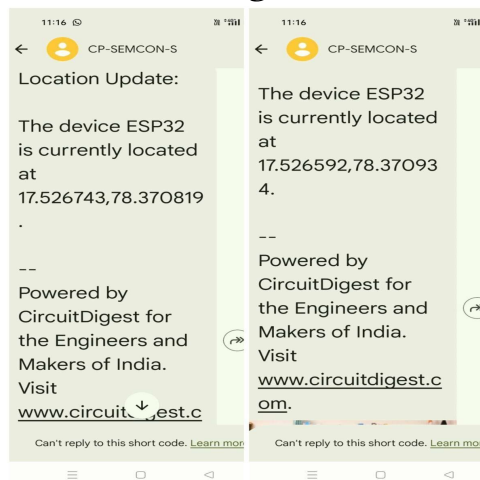


Fig.7



Fig.8

- GPS location successfully tracked
- Data sent to cloud in real time
- Distance calculated correctly
- Alerts generated properly
- SMS notifications received

Discussion:

The system provides reliable real-time monitoring. It ensures patient safety by sending alerts during abnormal movement.

However, performance may be affected by GPS accuracy and internet connectivity.

9. ADVANTAGES AND LIMITATIONS

Advantages:

- Real-time tracking
- Low cost
- Easy to use
- Reliable alerts

Limitations:

- Requires internet
- GPS may not work well indoors
- Depends on Wi-Fi

10. APPLICATIONS

- Alzheimer's patient monitoring
- Child tracking systems
- Elderly safety monitoring
- Personal tracking devices

11. CONCLUSION AND FUTURE SCOPE

Conclusion:

This project presents a real-time monitoring system for Alzheimer's patients using ESP32 and a Neo-6M GPS module. The system tracks the patient's location and sends the data to a cloud platform through Wi-Fi, allowing caregivers to monitor movements through a web application.

The geofencing feature helps in detecting when the patient moves outside a defined safe zone and triggers alerts such as SMS notifications. This improves patient safety and enables quick response from caregivers.

The system is low-cost, easy to use, and reliable for real-time tracking applications. Overall, it successfully demonstrates the use of IoT technology in healthcare monitoring.

Future Scope:

- Mobile app development
- Improved GPS accuracy
- Multiple safe zones
- Integration with wearable devices

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